



Review

Blood banking: The situation in India

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1. Introduction

India is the world's seventh-largest nation with a staggering population of over 1.4 billion as of July 2023. The Blood Transfusion Services (BTS) in India comprises of a sprawling network of approximately 4000 blood centres across the country [1]. Predominantly these are hospital-affiliated blood centres (77%) and including government (44%) and privately (33%) owned, rest 23% are classified as private charitable establishments, reflecting the diverse management landscape [2].

BTS are under the purview of the Directorate General of Health Services (DteGHS), which is under the jurisdiction of the Ministry of Health and Family Welfare (MOHFW). The Drug Controller General of India (DCGI) and the Blood Cell, National Health Mission (NHM) are coordinating its diverse functions.

Under the aegis of DteGHS, the National Blood Transfusion Council (NBTC) is the policymaking body and responsible for all aspects of organising, operating, and setting standards for a sustainable and secure BTS in India. Mandatory licensing of blood centres in India was implemented in 1997 by classifying blood as drugs in accordance to the Indian Pharmacopoeia. Blood centres are thus required to obtain a manufacturing license under the Drugs and Cosmetics Act (D and C Act) of 1940 and the Drugs and Cosmetics Rules of 1945, subjected to periodic amendments. The Central Drug Standard Control Organisation (CDSCO) headed by Drug Controller General of India (DCGI), and State Drug Control Organizations (headed by Drug controlled, State) play pivotal roles in approving licences and enforcing standardised transfusion practices.

At state level, the State Blood Transfusion Council (SBTC) and the Blood Safety Division of the State AIDS Control Society (SACS) coordinate all activities related to BTS and blood safety (Fig. 1). These entities oversee blood centres and its satellite (blood storage) centres, which are subject to regulation by the DCGI and State Drug Controller (DC, State). This multifaceted regulatory framework ensures the integrity, safety, and accessibility of blood transfusion services across India, safeguarding the health and well-being of its vast population.

A National Blood Policy was formulated by MOHFW in 2002, for an integrated strategy to ensure easy accessibility and adequate supply of safe and quality blood and blood components for all regardless of economic or social status. The policy emphasises on the quality, safety and

efficacy of blood and blood products, well-equipped blood centres with adequate infrastructure and trained manpower for its operation [4]. Additionally, in 2014, MOHFW also formulated a National Plasma Policy for ensuring abundant, easily accessible, and top-quality human plasma-derived proteins for therapeutic purposes [5].

The situation of the blood banking in India can be explained in the following headings:

2. Blood donor and collection

India collects approximately 12 million whole blood units annually, of which 70% are collected from voluntary blood donors and the rest comes from family and friends (replacement donors) [6]. Despite this, a shortfall of approximately 2 million units exists, falling short of the WHO-recommended 1% blood donation rate per population [7]. Since most of the blood centres are hospital-based blood centres (HBBC), the majority of the blood donations are performed within the HBBC. Collection capabilities also vary significantly among HBBC ranging from 5000 to 80,000 units per year. Over the past decade, efforts to increase the voluntary blood donation has resulted in an increased number of outdoor blood drives. For the same, MOHFW also allowed to private blood centres to organise voluntary blood donation drives in 2016. Although voluntary blood donation has increased significantly, smaller cities still rely on replacement donations for their blood needs, thus highlighting the regional disparities in blood donation practices within the country.

3. Blood components

Introduction of plastic blood bags in the early 1980 s in India, revolutionised the transfusion therapy by enabling the separation of whole blood into its constituent parts: red cells, platelets, and plasma components. During the initial phases, component services were largely tailored to meet the patient needs, but over the past decade, the number of facilities with component separation capabilities has grown substantially, thus allowing for a more rational approach for catering to the patients transfusion requirements. Currently more than 51% of the blood centres now have such services, separating 53% of the country's annual blood collection [8].

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